20-S1-Q1

Q(a) two point

uniform

vs

one point

(b) universal selection

vs

roulett wheel

(c) UA

Solution (a) ① two - point Example

P1 11|00|0

P2 00|11|1

C1 11110

C2 00001

② unifor crossover Example

P1 1|1|0|0|0

P2 0|0|1|1|1

C1 10010

C2 01101

③ preferred

(1) performance with 1 - point crossover depends on the order that variables occur in the representation

(2) Can never keep together genes from opposite end of string

(3) This is know as Positional Bias

(4) this is not usually the case

(b) Universal selection

① length = $\sum$ fitness

② N = population size

③ Place N equally spaced pointer a distance $F = \frac{\text{fitness}}{N}$ apart. The first pointer's origin is a single random number r drawn uniformly from $[0, F)$

④ whichever individuals the pointers land on are selected

Advantage

① SUS gives much lower sampling variance

② SUS keep diversity

③ roulette wheel over sample very fit individuals, causing permature convergenze

④ roulette whells occasionally miss moderate fitnesc individuals altogether

(c) ① $H = 1****$

order $O(H) = 1$

length $\delta(H) = 0$

$N = 4$

$L = 5$

$p_c = 0.8$

$p_m = 0.2$

②

String	fitness	Matches H
10001	15	✓
00101	5	×
01110	10	×
10010	20	✓

③ So $m(H, t) = 2$

$\bar{f} = (15 + 5 + 10 + 20)/4 = 12.5$

$f_H = (15 + 20)/2 = 17.5$

Goldberg's schema theorem

$$E[m(H, t+1)] = m(H, t) \frac{f_H}{\bar{f}} \left(1 - p_c \frac{\delta(H)}{L-1}\right)(1 - p_m)^{O(H)}$$

$$= 2 \times \frac{17.5}{12.5} \times 1 \times 0.8$$

$$= 2.24$$